Quantum Gates in Quantum Computing

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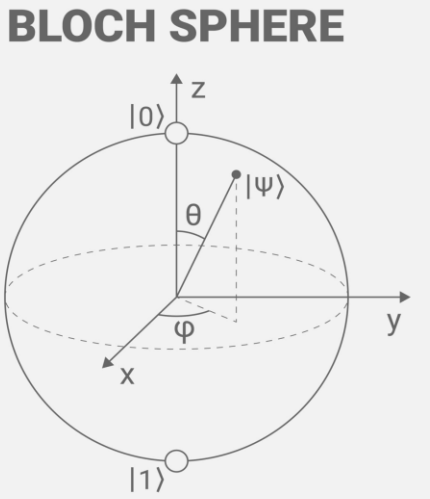
What is Quantum Computing?

Quantum computing is a computing paradigm that uses **qubits** and principles of **quantum mechanics** to process information.

Classical Computing	Quantum Computing
Uses bits	Uses qubits
Bit value = 0 or 1	Can be 0, 1, or both (superposition) $\lvert\psi\rangle=\alpha\lvert 0\rangle+\beta\lvert 1\rangle$
Sequential processing	Can process many possibilities at once



In quantum mechanics and computing, the Bloch sphere is a geometrical representation of the pure state space of a two-level quantum mechanical system (qubit), named after the physicist Felix Bloch.

What are Quantum Gates?

Quantum gates are the fundamental operations used to manipulate qubits in a quantum computer.

They are the quantum equivalent of classical logic gates such as AND, OR, and NOT.

Why Do We Need Quantum Gates?

Role of Quantum Gates

- Manipulate qubit states
- Perform quantum operations
- Build quantum circuits
- Enable quantum algorithms

In classical computing, logic gates like AND and OR manipulate bits.
Similarly, in quantum computing, quantum gates manipulate qubits.

However, instead of simply flipping between 0 and 1, quantum gates transform probability amplitudes using unitary matrices.

Properties of Quantum Gates

1. Unitarity Property

All quantum gates are represented by **unitary matrices**

Mathematical condition:

$$U^{\dagger}U = UU^{\dagger} = I$$

Where

$U^{\dagger}$ =conjugate transpose of U

Implications:

- Preserves probability amplitudes
- Ensures normalization of quantum states
- No information loss during computation.

Properties of Quantum Gates

2. Linearity

Quantum gates follow the **principle of linearity**.
If a quantum state is

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

Then

$$U\,|\psi\rangle = \alpha U\,|0\rangle + \beta U\,|1\rangle$$

Implications:

- Enables **superposition**
- Enables **quantum interference**
- Allows quantum parallelism

Properties of Quantum Gates

3. Composability

Quantum gates can be **combined sequentially** to form quantum circuits.
If two gates are applied:

$$U_1, U_2$$

The total operation becomes

$$U = U_2 U_1$$

Properties:

- Combination of unitary gates remains unitary
- Complex quantum algorithms are built from simple gates

Properties of Quantum Gates

4. Reversibility

Reversibility means that the **original input state can always be recovered from the output state**.

In quantum computing:

- Every quantum gate is **reversible**
- No information is destroyed during computation
- The input state can be obtained by applying the **inverse operation**

Mathematically:

$$U^{-1}U = I$$

Where

U= quantum gate, U^{-1} =inverse gate , I= identity operation

This means applying a gate and then its inverse returns the **original quantum state**.

Difference Between Classical and Quantum Gates

Classical Gates	Quantum Gates
Often irreversible	Always reversible
Information may be lost	Information is preserved
Example: AND gate	Example: Pauli-X gate

Example: Pauli-X Gate

Matrix of Pauli-X gate:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Transformation:

$$X \mid 0\rangle = \mid 1\rangle$$
$$X \mid 1\rangle = \mid 0\rangle$$

Applying the gate twice:

$$X(X \mid 0\rangle) = \mid 0\rangle$$
$$X(X \mid 1\rangle) = \mid 1\rangle$$

Therefore

$$X^2 = I$$

This shows that the **X gate is its own inverse**.

Example: Hadamard Gate

Matrix:

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Transformations:

$$H \mid 0\rangle = \frac{\mid 0\rangle + \mid 1\rangle}{\sqrt{2}}$$

$$H \mid 1\rangle = \frac{\mid 0\rangle - \mid 1\rangle}{\sqrt{2}}$$

Applying H again:

$$H(H \mid 0\rangle) = \mid 0\rangle$$

$$H(H \mid 1\rangle) = \mid 1\rangle$$

Thus

$$H^2 = I$$

So Hadamard gate is also **reversible**.

Question

Exercise 2.22. Consider a map U that transforms the Z-basis states as follows:

$$U|0\rangle = |0\rangle + |1\rangle,$$
$$U|1\rangle = |0\rangle - |1\rangle.$$

Say $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is a normalized quantum state, i.e., $|\alpha|^2 + |\beta|^2 = 1$.

(a) Calculate $U|\psi\rangle$.

(b) From your answer to (a), is U a valid quantum gate? Explain your reasoning.

(a) Calculation of $U|\psi\rangle$

Step 1: Apply linearity of quantum operators

Quantum operators are **linear**, therefore

$$U(\alpha|0\rangle + \beta|1\rangle) = \alpha U|0\rangle + \beta U|1\rangle$$

Step 2: Substitute the given transformations

$$U|\psi\rangle = \alpha(|0\rangle + |1\rangle) + \beta(|0\rangle - |1\rangle)$$

Step 3: Expand the expression

$$U|\psi\rangle = \alpha|0\rangle + \alpha|1\rangle + \beta|0\rangle - \beta|1\rangle$$

Step 4: Group the basis states

$$U|\psi\rangle = (\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle$$

Final Result

$$\boxed{U|\psi\rangle = (\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle}$$

(b) Is U a Valid Quantum Gate?

A **valid quantum gate must be unitary**, meaning it must **preserve normalization** of the quantum state.

That is:

$$\langle \psi' | \psi' \rangle = 1$$

where

$$|\psi'\rangle = U|\psi\rangle$$

Step 1: Compute the norm of the new state

$$|\psi'\rangle = (\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle$$

The norm becomes

$$\langle \psi' | \psi' \rangle = |\alpha + \beta|^2 + |\alpha - \beta|^2$$

Step 2: Expand both terms

$$|\alpha + \beta|^2 = |\alpha|^2 + |\beta|^2 + \alpha\beta^* + \alpha^*\beta$$

$$|\alpha - \beta|^2 = |\alpha|^2 + |\beta|^2 - \alpha\beta^* - \alpha^*\beta$$

Step 3: Add the expressions

$$|\alpha+\beta|^2+|\alpha-\beta|^2=2|\alpha|^2+2|\beta|^2$$

Factor:

$$=2(|\alpha|^2+|\beta|^2)$$

Using normalization condition:

$$|\alpha|^2+|\beta|^2=1$$

Therefore

$$\langle\psi'|\psi'\rangle=2$$

Step 4: Interpretation

The resulting state has norm:

$$\sqrt{2}$$

instead of **1**, meaning the transformation **does not preserve normalization**.

Therefore it is **not unitary**.

Matrix Representation

From the basis transformations:

$$U = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Check unitarity:

$$U^\dagger U = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

Since

$$U^\dagger U \neq I$$

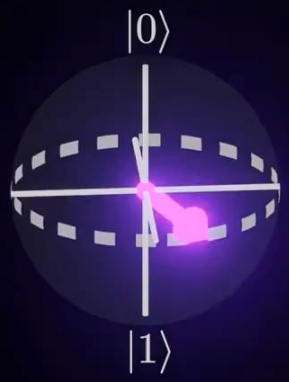
the operator is **not unitary**.

Final Conclusion

$$\boxed{U|\psi\rangle = (\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle}$$

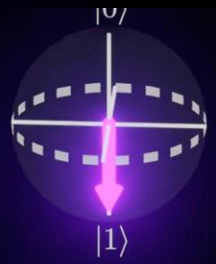
$$\boxed{U \text{ is NOT a valid quantum gate because it is not unitary and does not preserve normalization.}}$$

Bloch Sphere



100% $|0\rangle$

0% $|1\rangle$



0% $|0\rangle$

100% $|1\rangle$

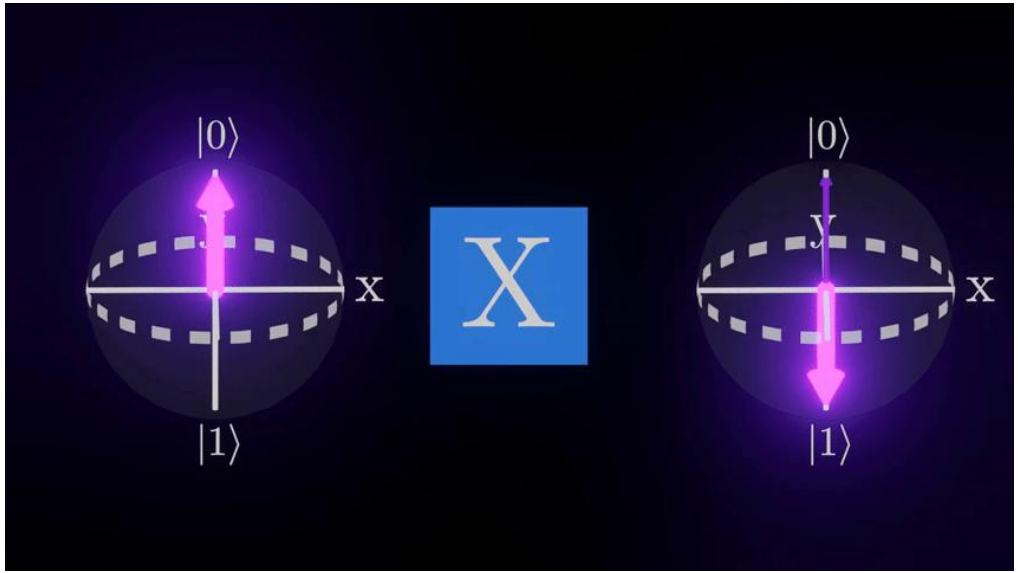
Single Qubit Gates: Pauli-X Gate (σ_x)

Pauli-X (Bit Flip)

Matrix: $$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Also written as: σx

Circuit symbol: ⊕ (NOT symbol in quantum circuits)



Rotation Angle $X = e^{-i\frac{\pi}{2}\sigma_x}$

Equivalent to:
Rotation by **π radians (180°)** around X-axis on Bloch sphere.

Action on Basis States

X|0⟩= |1⟩
X|1⟩= |0⟩

Acts like classical NOT gate.

Single Qubit Gates: Pauli-Y Gate (σ_γ)

Pauli-Y (Bit + Phase Flip)

Matrix: $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

Also written as: σy

Rotation Angle $Y = e^{-i\frac{\pi}{2}\sigma_y}$

Equivalent to:
Rotation by **π radians (180°)** around Y-axis on Bloch sphere.

Action on Basis States

Y|0⟩= i |1⟩
Y|1⟩= −i |0⟩

Flips state + adds phase factor.

Single Qubit Gates: Pauli-Z Gate (σ_z)

Pauli-Z (Phase Flip)

Matrix: $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Also written as: σz

Rotation Angle $Z = e^{-i\frac{\pi}{2}\sigma_z}$

Equivalent to:
Rotation by **π radians (180°)**
around Z-axis on Bloch sphere.

Action on Basis States

$Z|0\rangle = |0\rangle$
$Z|1\rangle = -|1\rangle$

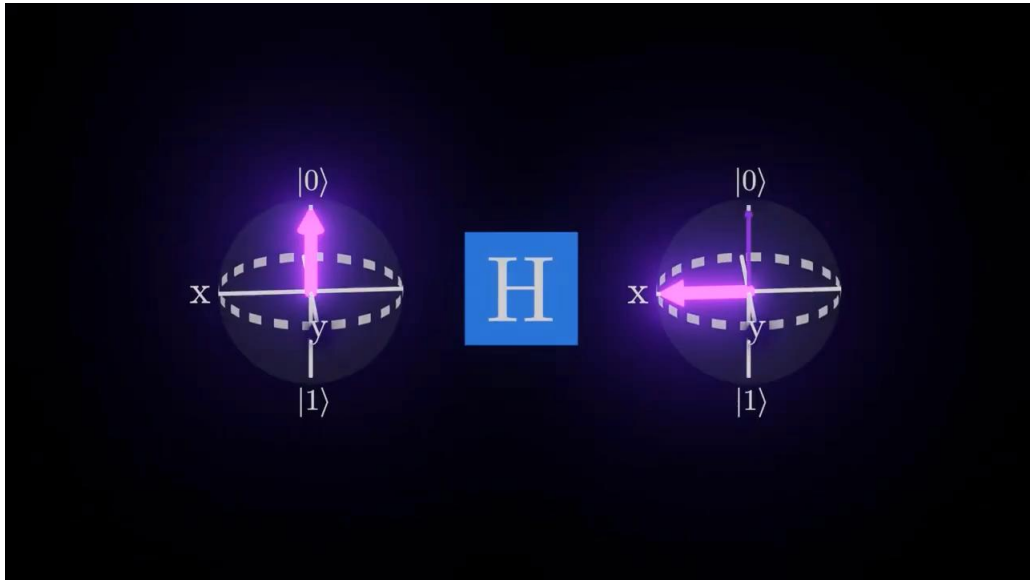
This is a **phase flip** gate.(Changes phase, not value)

Single Qubit Gates: Hadamard Gate (H)

The **Hadamard gate (H)** is one of the most important single-qubit quantum gates.

It is used to:

- Create superposition
- Enable quantum parallelism
- Convert between computational and diagonal basis
- Initiate interference in quantum algorithms



Matrix Representation

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Single Qubit Gates: Hadamard Gate (H)

Action on Basis States

On |0⟩:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

On |1⟩:

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Interpretation:

Creates equal superposition

Introduces relative phase difference

Enables interference

Rotation Interpretation

Although H is not purely a single-axis rotation, it can be expressed as:

$$H = \frac{X + Z}{\sqrt{2}}$$

Geometric interpretation:

Equivalent to rotation by **π radians (180°)** about axis:

$$\frac{1}{\sqrt{2}}(X + Z)$$

This axis lies halfway between X and Z axes on Bloch sphere.

Single Qubit Gates: Hadamard Gate (H)

Hadamard gate is essential in:

1. Quantum Parallelism

Applying H to n qubits:

Creates uniform superposition of all basis states.

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle$$

2. Shor's Algorithm

Used in quantum Fourier transform initialization.

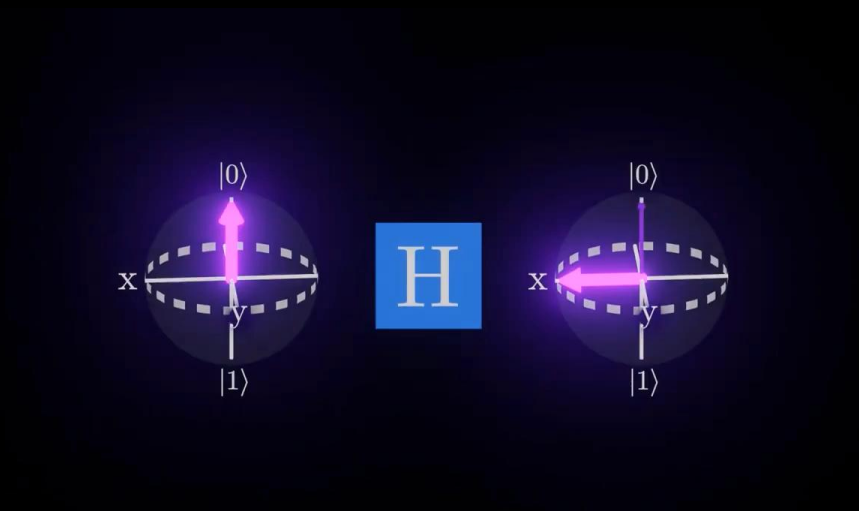
3. Grover's Algorithm

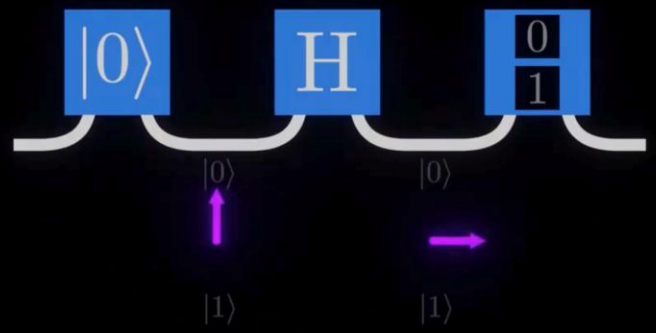
Used in diffusion operator.

4. Quantum Teleportation

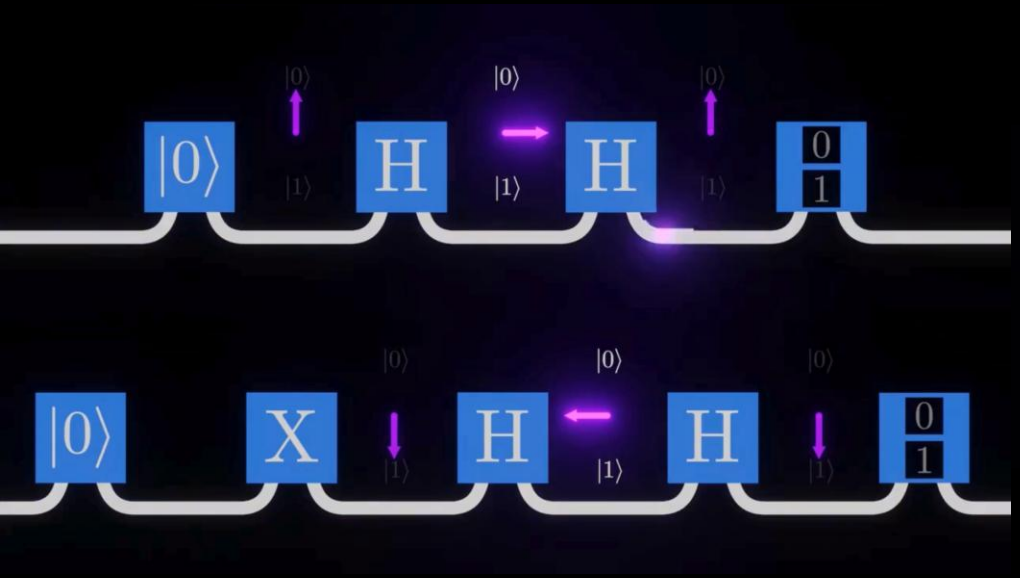
Used in Bell state preparation.

Single Qubit Gates: Hadamard Gate (H)

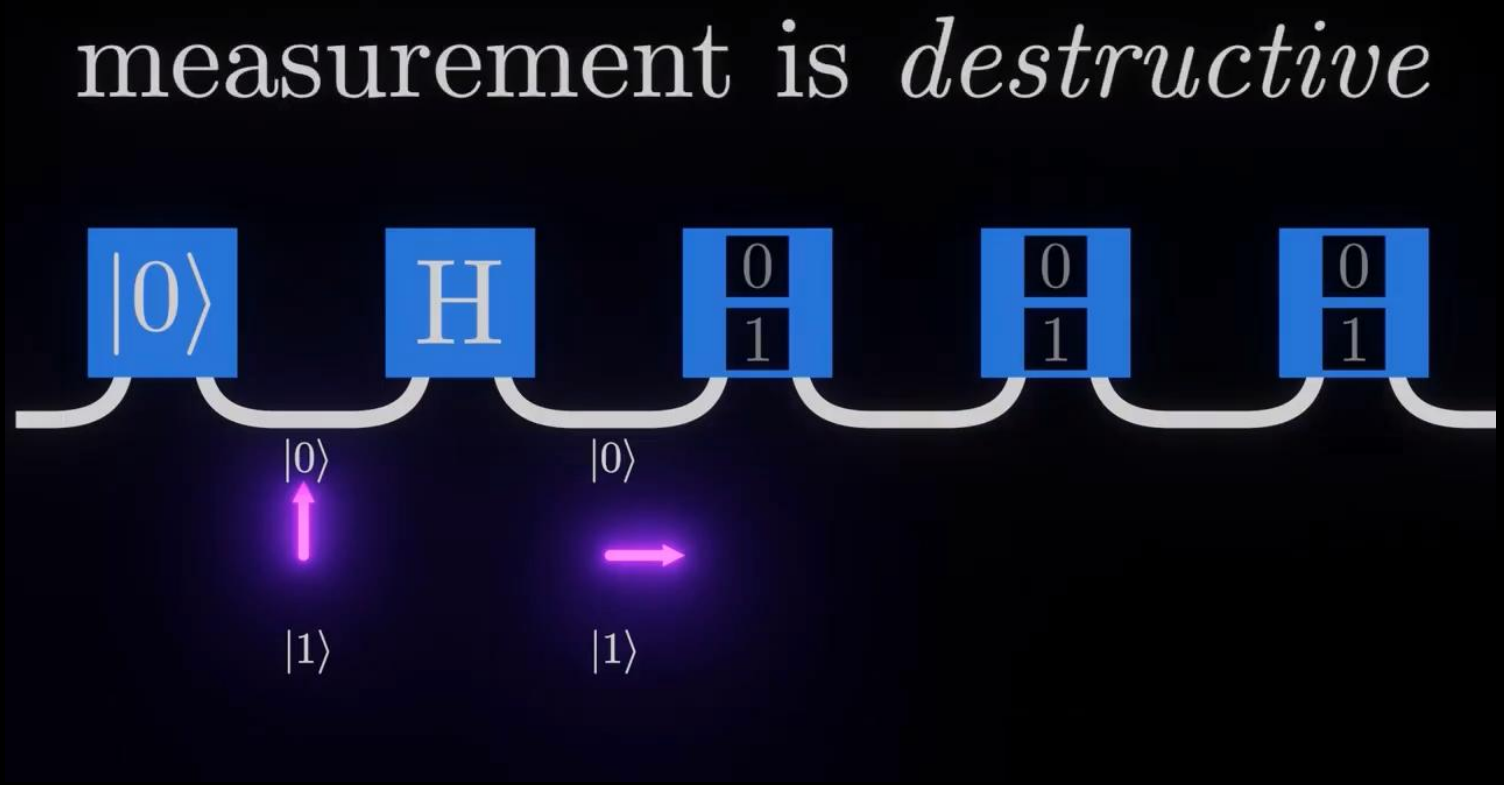




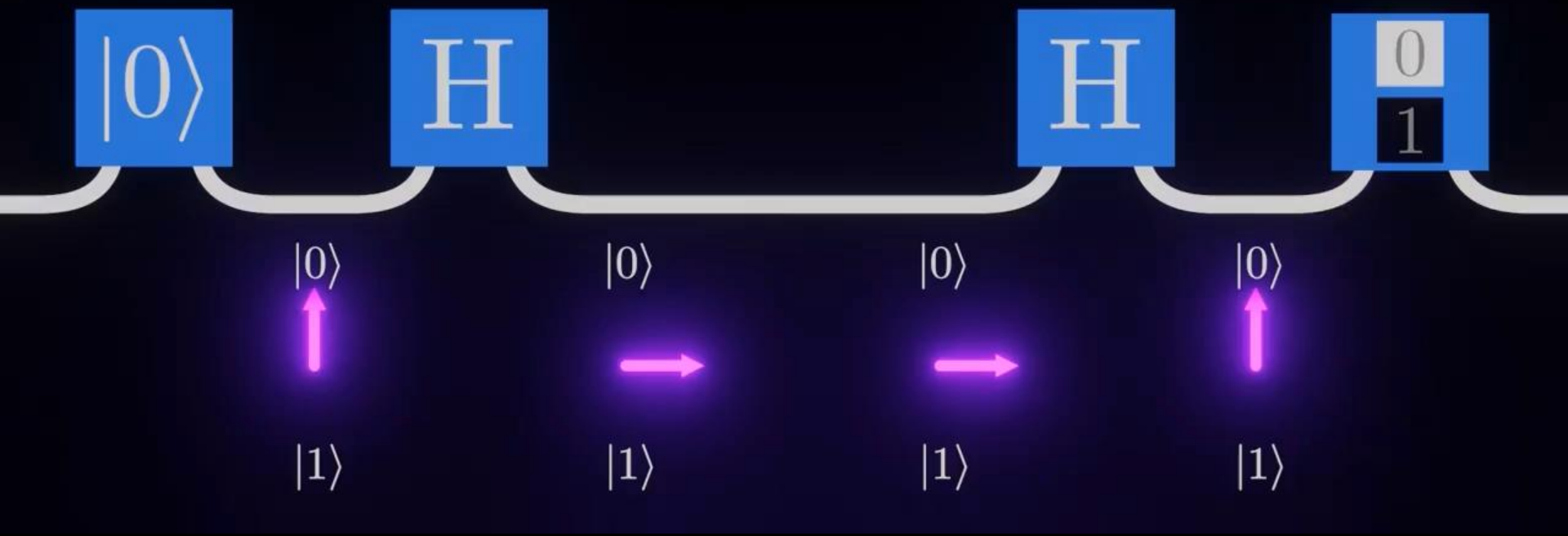
Single Qubit Gates: Hadamard Gate (H)



Collapsing a Superposition



Putting a Measurement Gate b/w two H-Gates



Single Qubit Gates: **Phase Gates (S and T)**

The S Gate (The Phase Gate)

Definition: Often called the √Z gate because applying it twice results in a Pauli-Z gate.

Action: Rotates the qubit state by $\pi/2$ **(90°)** around the Z-axis on the Bloch Sphere.

Effect on States:

$|0\rangle$ to $|0\rangle$ (No change)

$|1\rangle$ to $i|1\rangle$ (Adds a imaginary phase)

The Matrix:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

Single Qubit Gates: **Phase Gates (S and T)**

The T Gate (The $\pi/8$ Gate)

Definition: A fundamental "non-Clifford" gate. It is the most critical gate for achieving **Universal Quantum Computation**.

Known as the "**$\pi/8$** gate" (due to a different mathematical convention).

Action: Rotates the qubit state by **$\pi/4$ (45°)** around the Z-axis.

Effect on States:

$|0\rangle$ to $|0\rangle$

$|1\rangle$ to $e^{i\pi/4}|1\rangle$

The Matrix:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Single Qubit Gates:**Rotation Gates ($R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$)**

Explanation: These are generalized gates that allow rotation by *any* arbitrary angle θ

General Formula (R_z example):

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

Key Features

Continuous parameter
Infinite possible gates
Generate all single-qubit unitaries

Use Case: Essential for fine-tuning quantum algorithms like VQE or QAOA.

Multi-Qubit Gates: The CNOT Gate

Full Name: Controlled-NOT Gate.

Logic: Uses two qubits (Control and Target). If the Control is $|1\rangle$, it flips the Target.

Significance: This gate creates **Entanglement**.

Symbol: A small dot (control) connected by a vertical line to an X-gate (target).

The Matrix:

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Multi-Qubit Gates: (CZ and SWAP)

Full Name: CZ (Controlled-Z)

Logic: Flips the phase of the target qubit only if the control qubit is $|1\rangle$

Significance: Applies Z gate if both qubits = 1.

Key Feature: Phase entangling gate.

The Matrix:

$$CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Multi-Qubit Gates: (CZ and SWAP)

Full Name: SWAP Gate

Logic: Exchanges the states of two qubits.

Significance: Useful for moving information across a quantum chip where qubits aren't all connected.

Visual: Show the SWAP symbol (two "x" marks connected by a line).

The Matrix:

$$SWAP = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multi Qubit State: Toffoli Gate (CCNOT – Controlled Controlled NOT)

Definition: The **Toffoli gate** is a **three-qubit reversible quantum gate** that flips the third (target) qubit **only if both control qubits are 1**.

Also called: CCNOTOT

Operation Rule:

Let qubits be:
$|a,b,c\rangle$ Then: $|a,b,c\rangle \rightarrow |\, a,b,c \oplus (a \cdot b)\,\rangle$
Meaning:

If a=1 AND b=1 → flip c

Otherwise → no change

It swaps only the last two states: $|110\rangle \leftrightarrow |111\rangle$

Matrix Representation:

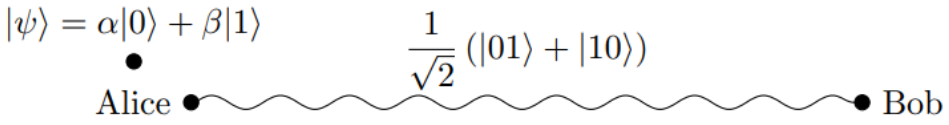
$$\mathrm{CCNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Common Quantum Gates with Circuit Symbols and Matrix Representations

Operator	Gate(s)		Matrix
Pauli-X (X)	X	$\oplus$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)	Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)	Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)	H		$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)	S		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)	T		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)	$\bullet$ $\oplus$		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)	$\bullet$ Z	$\bullet$ $\bullet$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP	$\times$	$\times$ $\times$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)	$\bullet$ $\bullet$ $\oplus$		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

Question

Exercise 6.16. Alice wants to teleport a qubit in an unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Bob. Instead of sharing two entangled qubits in the $|\Phi^+\rangle$ state, they share two entangled qubits in the $|\Psi^+\rangle$ state:



Altogether, the initial state of the system is

$$\begin{aligned}|\psi\rangle|\Psi^+\rangle &= (\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ &= \frac{1}{\sqrt{2}}(\alpha|001\rangle + \alpha|010\rangle + \beta|101\rangle + \beta|110\rangle).\end{aligned}$$

So, the left two qubits are Alice's, and the right qubit is Bob's.

(a) Show that if Alice applies CNOT to her two qubits, followed by H to her left qubit, the state of the system becomes

$$\begin{aligned}\frac{1}{2}\big[&|00\rangle(\beta|0\rangle + \alpha|1\rangle) + |01\rangle(\alpha|0\rangle + \beta|1\rangle) \\ &+ |10\rangle(-\beta|0\rangle + \alpha|1\rangle) + |11\rangle(\alpha|0\rangle - \beta|1\rangle)\big].\end{aligned}$$

(b) Next, Alice measures both of her qubits. What values can she get, with what probabilities, and what does the state collapse to in each case?

(c) Finally, Alice tells Bob the results of her measurement. For each possible result, what should Bob do to his qubit so that it is $\alpha|0\rangle + \beta|1\rangle$, the state that Alice wanted to teleport to him?

Given initial state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Shared state:

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

Total state:

$$|\psi\rangle|\Psi^+\rangle = \frac{1}{\sqrt{2}}(\alpha|001\rangle + \alpha|010\rangle + \beta|101\rangle + \beta|110\rangle)$$

Left two qubits = Alice
Right qubit = Bob

◆ Step 1 — Apply CNOT (first → second qubit)

Rule:
If first bit = 1 → flip second bit.

Apply to each term:

- $|001\rangle$ → stays same
- $|010\rangle$ → stays same
- $|101\rangle$ → becomes $|111\rangle$
- $|110\rangle$ → becomes $|100\rangle$

So state becomes:

$$\frac{1}{\sqrt{2}}(\alpha|001\rangle + \alpha|010\rangle + \beta|111\rangle + \beta|100\rangle)$$

Step 2 — Apply Hadamard to first qubit

Recall:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Apply to each term separately and simplify.

After expanding and grouping by first two qubits, we obtain:

$$\boxed{\frac{1}{2}\Big[|00\rangle(\beta|0\rangle + \alpha|1\rangle) + |01\rangle(\alpha|0\rangle + \beta|1\rangle) + |10\rangle(-\beta|0\rangle + \alpha|1\rangle) + |11\rangle(\alpha|0\rangle - \beta|1\rangle)\Big]}$$

🔵 Part (b) Question

> Alice measures both of her qubits.
> What values can she get, with what probabilities,
> and what does the state collapse to?

✅ Step 1 — What values can she get?

She measures 2 qubits.

Each qubit can be 0 or 1.

So possible outcomes:

00
01
10
11

✅ Step 2 — What are the probabilities?

Each term has coefficient $\frac{1}{2}$

Probability = square of coefficient

$$(1/2)^2 = 1/4$$

So:

$$P(00) = P(01) = P(10) = P(11) = \frac{1}{4}$$

Each outcome has equal probability.

✅ Step 3 — What does system collapse to?

When Alice measures, only the corresponding bracket survives.

If Alice measures 00

System collapses to:

$$\beta|0\rangle + \alpha|1\rangle$$

If Alice measures 01

System collapses to:

$$\alpha|0\rangle + \beta|1\rangle$$

If Alice measures 10

System collapses to:

$$-\beta|0\rangle + \alpha|1\rangle$$

(Global minus irrelevant.)

If Alice measures 11

System collapses to:

$$\alpha|0\rangle - \beta|1\rangle$$

Possible outcomes: 00, 01, 10, 11

Each with probability $\frac{1}{4}$

Collapsed states:

00 $\rightarrow \beta|0\rangle + \alpha|1\rangle$

01 $\rightarrow \alpha|0\rangle + \beta|1\rangle$

10 $\rightarrow -\beta|0\rangle + \alpha|1\rangle$

11 $\rightarrow \alpha|0\rangle - \beta|1\rangle$

What part (c) asks

For each possible measurement result, what should Bob do so that his qubit becomes $\alpha|0\rangle + \beta|1\rangle$?

So:

1. Look at Bob's state from part (b).
2. Compare with original state.
3. Decide correction.

Original state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

🔵 Case 1 — Alice measures 00

Bob has:

$$\beta|0\rangle + \alpha|1\rangle$$

Compare structure:

$|0\rangle$ and $|1\rangle$ are swapped.

That's what **X gate** does.

So Bob applies:

$$\boxed{X}$$

🔵 Case 2 — Alice measures 01

Bob has:

$$\alpha|0\rangle + \beta|1\rangle$$

Exactly original state.

So Bob applies:

$$\boxed{I}$$

(No operation)

🔵 Case 3 — Alice measures 10

Bob has:

$$-\beta|0\rangle + \alpha|1\rangle$$

Ignore global minus:

$$\beta|0\rangle - \alpha|1\rangle$$

Structure = swapped + sign difference.

That corresponds to:

$$\boxed{XZ}$$

🔵 Case 4 — Alice measures 11

Bob has:

$$\alpha|0\rangle - \beta|1\rangle$$

Only sign on $|1\rangle$ is flipped.

That is exactly what Z does.

So Bob applies:

$$\boxed{Z}$$

Alice measures	Bob applies
00	X
01	I
10	XZ
11	Z

References

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Thank You!